National Higher School of Autonomous Systems Technology
Academic year : 2025/2026

Module : Physics 4
2nd year 2025/2026

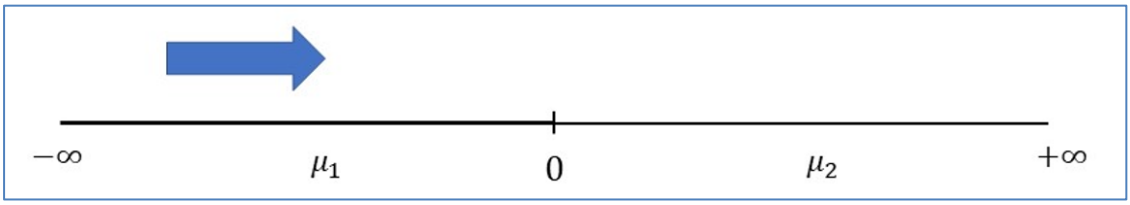
Set N⁰ 2
Vibrating string

Exercise 1:

1-A string of length 4.35 m and mass 137 g is under a tension of 125 N. A standing wave has formed which has seven nodes including the endpoints. What is the frequency of this wave? Which harmonic is it? What is the fundamental frequency?
-Represent the fundamental mode and the first three overtones
2-A string fixed at one end only, is vibrating in its ninth harmonic mode. The speed of a wave on the string is $v = 25.8\ m/s$ and the string has a length of 8.25 m. What is the frequency of this wave? What is the wavelength of the wave? What is the fundamental frequency?
-Represent the fundamental mode and the first three overtones

Exercise 2:

Two semi-infinite strings positioned along an $x'Ox$ axis are connected at x=0 . The string in the region x<0 has a linear mass density μ_1 . The string extending from 0 to $+\infty$ has a linear mass density μ_2= 0.25μ_1. An incident wave of amplitude U_0 and angular frequency ω arrives from -∞ and propagates in the direction of increasing x. At x=0, the wave undergoes reflection.



1. Calculate the reflection coefficient at x=0.

2. Show that the resulting wave in the region x<0 varies between two values U_{max} and U_{min}. Determine U_{max} and U_{min} , as well as the positions of the vibration maxima and minima. Calculate the standing wave ratio (SWR).

3. Demonstrate that this system is equivalent to a string terminated at x= 0 by a damper, and specify the value of the damping coefficient.

4. In the case where μ_1= μ_2:
- What happens to the reflection coefficient R (in amplitude) at x=0 ?

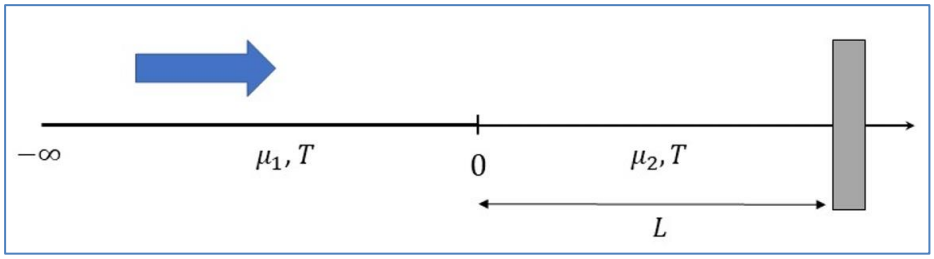
- Calculate the linear densities of kinetic and potential energy, e_c and e_p , at any point x along the $x'Ox$ axis.

- Deduce the total energy density and calculate its average value over a wavelength λ.

Exercise 3:

A semi-infinite string with a linear mass density μ_1 is stretched along the $x'Ox$ axis between $-\infty$ and x=0. At x=0, it is connected to a second string with a linear mass density μ_2 and length L. At x=L, this second string is attached to a rigid wall. The tension in both strings is denoted by T.

National Higher School of Autonomous Systems Technology
Academic year : 2024/2025



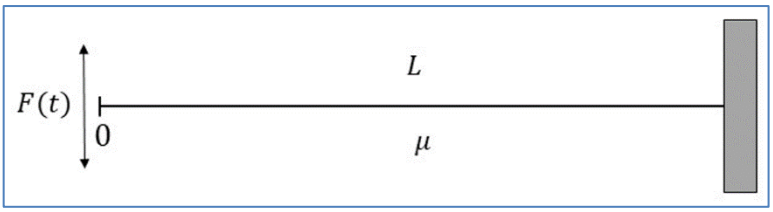
An incident wave with angular frequency ω is sent along the string, and at x=0, it undergoes reflection and transmission phenomena. We write the wave equations in the two regions of space:

$x < 0 : \quad u_1(x,t) = A_1 e^{j(\omega t - kx)} + B_1 e^{j(\omega t + kx)}$
$x > 0 : \; u_2(x,t) = A_2 e^{j(\omega t - k_2 x)} + B_2 e^{j(\omega t + k_2 x)}$

1- Write the continuity equation at $x=0$ and derive the relationship between A_1, A_2, B_1, and B_2
2- Write the condition at x=L and derive the ratio $\frac{B_2}{A_2}$.
3- Determine the reflection coefficient $R=\frac{B_1}{A_1}$ at $x=0$.
4- What relationship must exist between L_2 and λ_2 for a node to form at $x=0$?

Exercise 4:

A string of length L and linear mass density μ is stretched horizontally with a tension T. It is fixed at its end at $x = L$ to a rigid support. The end at $x = 0$ is subjected to a sinusoidal force with an amplitude F_0 and angular frequency w . The transverse displacement of a point at position x on the string at time t is denoted as $y(x,t)$.

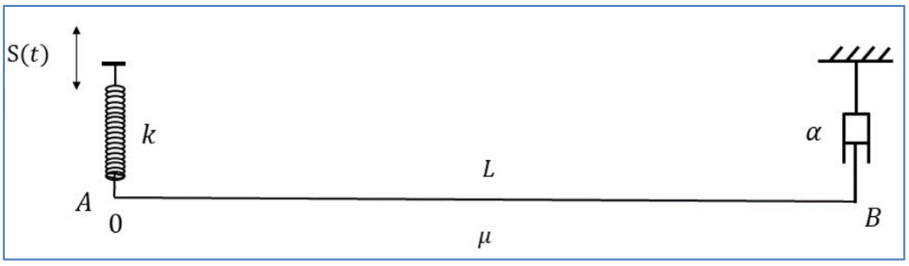


1- a. Write the general expression for $y(x,t)$.
   b. Provide the boundary conditions at $x=0$ and $x = L$.
   c. Show that the expression for $y(x,t)$. can be written in the form: $y(x,t)$. = f(x).g(t)
   d- Give the expression for f(x).in terms of F_0, the wave number k, L, and T.
2- Determine the positions of the points of maximum amplitude (antinodes) as a function of the wavelength λ and the length L of the string. What is the distance between two successive nodes?
3- For which angular frequencies is a resonance phenomenon observed?

Exercise 5:

A string of length $AB = L = 2m$ and mass $m = 80g$ is subjected to a tension T. Its end A is connected to a spring with stiffness constant k positioned vertically.
The other end of the spring is subjected to a vertical sinusoidal displacement with angular frequency $w=100\pi$ and amplitude s_0 $s\ t = s_0 e_{jwt}$ At $x = L$, the string is connected to a damper with a viscous friction coefficient $\alpha=0.2N/m$ (see figure).

National Higher School of Autonomous Systems Technology
Academic year : 2024/2025



The tension T of the string is adjusted so that there is no reflection at point B. The string vibrates in the vertical plane along the Oy axis. The displacement of a point at position x on the string is represented by its position $y(x,t)$.

1. Justify without calculation that the displacement is written as: $y(x,t) = y_0 e^{j(\omega\left(t-\frac{x}{v}\right)+\varphi)}$
2. Calculate the reflection coefficient at $x = L$. Deduce the values of the string tension T and the phase velocity v.
3. Provide the expression of the displacement $y\ x,t$ for the point at position x on the string. Deduce that the points at x=0 and x=L vibrate in phase.
4. What is the value of the impedance at point A? Determine the amplitude y and the phase φ of the vibration at point A. Deduce the expression of the displacement $y\ x,t$ for a point at position x. In which case is the phase zero? What is the value of y_0 in that case?

Exercise 6:

The system shown in the figure below consists of a cylindrical pulley of mass M and radius R . At two diametrically opposite points A and B , located in a horizontal plane at equilibrium, a spring of stiffness β and a string of linear mass density μ are fixed. The string is stretched under a tension T . At point C , located on the surface of the pulley and perpendicular to AB , a tangential force of small amplitude $F(t) = F_0 cos(w\ t)$ is applied. At the other end of the string, a damper with a viscous friction coefficient α is fixed, chosen so that the string supports a progressive sinusoidal wave.

1. Calculate the impedance $Z(x)$ at a point x on the string.
2. What is the input impedance of the string at $x = 0$?
3. Calculate the average power P supplied by the mechanical system to the string.
4. Knowing that at a frequency of oscillation of 10 Hz, P reaches its maximum value P_{max} , calculate: the stiffness constant β and the linear mass density μ.

Given: $M = 0.5\ kg$,, $T = 10\ N$.